Self-Supervised Masked Image Modeling on High-Resolution Multispectral Aerial Imagery for Automated Agricultural Infrastructure Detection

First Author[a], Second Author[a]

[a] *Department of Agricultural Engineering, University Name, Address Line, City, 00000, State, Country*

Abstract

Monitoring agricultural production facilities, such as poultry farms and greenhouses, is essential for economic forecasting, biosecurity, and environmental compliance. Traditional supervised deep learning models for these tasks heavily rely on ImageNet pre-training, which introduces a substantial domain gap when applied to overhead, multispectral aerial imagery. In this study, we propose a domain-specific foundation model utilizing Masked Image Modeling (MIM) via the Spatial Masked Autoencoder (SparK) framework. We pre-trained a convolutional backbone entirely from scratch on a massive, unlabelled dataset comprising 1.5 Terabytes (approximately 10 million 512×512 tiles) of National Agriculture Imagery Program (NAIP) imagery. The learned representations were evaluated across three diverse downstream agricultural and remote sensing tasks: poultry farm detection, greenhouse detection (GRSD), and complex multi-class aerial detection (DOTA). Our results demonstrate that self-supervised pre-training on domain-specific unlabelled data achieves high parity with heavily supervised ImageNet baselines, obtaining 97.9% mAP50 on poultry farms, 96.7% on greenhouses, and 90.8% on the DOTA dataset. Furthermore, an ablation study on data efficiency (10% few-shot regime) revealed critical insights into the pre-training scale required for self-supervised models to generalize under data scarcity. This research confirms that leveraging abundant unannotated satellite imagery via self-supervised learning is a highly scalable and viable pathway for next-generation agricultural monitoring and precision agriculture workflows, removing the dependency on costly manual annotations.

Keywords: Agricultural Infrastructure, Self-Supervised Learning, Masked

Image Modeling, Remote Sensing, NAIP, Object Detection, Precision Agriculture

1. Introduction

The global expansion of intensive agricultural facilities, including confined animal feeding operations (CAFOs) such as poultry farms and expansive greenhouse complexes, necessitates robust monitoring frameworks. Accurate mapping of these structures is vital for mitigating environmental impacts, ensuring biosecurity during pathogen outbreaks (e.g., Avian Influenza), and facilitating regional economic planning. Historically, surveying such facilities relied on manual geographic information system (GIS) digitization, an approach that is labor-intensive, error-prone, and unscalable across national scales.

The advent of high-resolution aerial imagery, such as the National Agriculture Imagery Program (NAIP), combined with deep convolutional neural networks (CNNs), has revolutionized automated feature extraction. Current state-of-the-art architectures, such as the YOLO (You Only Look Once) series, excel at rapid object detection. However, they are traditionally initialized with weights pre-trained on ImageNet—a dataset consisting of 1.2 million explicitly labeled, natural, ground-level photographs. Applying ImageNet priors to top-down, multispectral remote sensing data introduces a significant "domain gap," forcing the network to relearn fundamental spatial hierarchies during fine-tuning.

Self-Supervised Learning (SSL), specifically Masked Image Modeling (MIM), presents a paradigm shift by enabling networks to learn robust visual representations without human-generated labels. By masking a portion of an image and training the network to reconstruct the missing pixels, the model inherently learns the structural and semantic composition of the data. While MIM has seen immense success in natural imagery, its application to massive-scale, high-resolution agricultural datasets remains underexplored.

In this paper, we present a comprehensive pipeline that applies the SparK (Spatial Masked Image Modeling) framework to pre-train a YOLO backbone entirely from scratch using 1.5 Terabytes of unlabelled NAIP aerial imagery. We fine-tune this domain-specific foundation model on three distinct downstream tasks to evaluate its representational transferability. We hypothesized that an SSL model pre-trained exclusively on unlabelled aerial

data can match or exceed the performance of a fully supervised ImageNet baseline, thereby removing the dependency on costly manual annotations in agricultural remote sensing.

2. Related Work

2.1. Deep Learning in Agricultural Remote Sensing

The application of convolutional neural networks (CNNs) in precision agriculture has seen exponential growth. Early applications focused on pixel-wise segmentation of crop types using Landsat or Sentinel-2 data. With the availability of sub-meter imagery like NAIP, research has shifted towards localized object detection, such as identifying individual trees, tractors, and agricultural infrastructure. Studies utilizing Faster R-CNN and YOLO architectures have proven highly effective for detecting greenhouses and CAFOs. However, nearly all these studies initialize their models with weights pre-trained on ImageNet, inheriting a visual prior that is fundamentally misaligned with overhead, top-down perspectives.

2.2. Self-Supervised Learning and Masked Image Modeling

Self-Supervised Learning (SSL) has emerged as a powerful alternative to supervised pre-training. Contrastive learning methods (e.g., SimCLR, MoCo) initially bridged the gap, but they rely heavily on domain-specific data augmentations. Recently, Masked Image Modeling (MIM), popularized by the Masked Autoencoder (MAE), has demonstrated superior scalability. MAE operates on Vision Transformers (ViTs) by masking up to 75% of image patches and reconstructing them. Extending this to convolutional networks, the SparK framework utilizes sparse convolutions to perform hierarchical MIM, making it highly suitable for architectures like ResNet and YOLO. This study builds upon these foundational works by applying convolutional MIM to a massive agricultural remote sensing dataset.

3. Materials and Methods

3.1. Datasets

3.1.1. SSL Pre-training Dataset (NAIP)

To eliminate the domain gap, we compiled a massive pre-training dataset comprising 9.87 million high-resolution tiles (512×512 pixels) derived from NAIP imagery (sourced from Satlas and SpaceNet). Totaling approximately

1.5 Terabytes, this dataset encompasses diverse geographic topologies across the United States. Unlike standard computer vision datasets, NAIP inherently contains multispectral data, incorporating True Color (TCI) RGB and Near-Infrared (IR) bands. The IR band is critical for agricultural applications, as it provides distinct spectral signatures that differentiate artificial structures from surrounding vegetation.

3.1.2. Downstream Fine-Tuning Datasets

To rigorously evaluate transferability, we utilized three structurally distinct datasets (summarized in Table 1):

1. **Poultry Farm Dataset:** A specialized dataset containing precise oriented bounding boxes (OBB) differentiating *Broiler Houses* from *Broiler Breeders*.
2. **Greenhouse Remote Sensing Dataset (GRSD):** A dataset comprising 2,101 images of complex, reflective, plastic greenhouse structures.
3. **DOTA Benchmark:** A large-scale, 15-class dataset featuring densely packed, highly rotated aerial objects (e.g., ships, planes, storage tanks), serving as a benchmark for generalized aerial understanding.

Table 1: Characteristics of the pre-training and downstream evaluation datasets.

Dataset	**Role**	**Images**	**Resolution**	**Classes**
NAIP Subset	Pre-training	$\sim$9.87M	512×512	N/A (Unlabelled)
Poultry Farm	Fine-Tuning	1,423	Varies	Broiler House, Breeder
GRSD	Fine-Tuning	2,101	Varies	Greenhouse
DOTA	Fine-Tuning	2,806	Varies	15 diverse aerial classes

3.2. Proposed Architecture

3.2.1. Masked Image Modeling (SparK)

We adopted the SparK framework for pre-training. Given an input image X, we mask a subset of patches to generate a masked image X_M. The encoder E processes the visible patches, and a lightweight decoder D reconstructs the

original image. The objective is to minimize the Mean Squared Error (MSE) between the reconstructed pixels $\hat{X}$ and the original pixels X in the masked regions:

$$\mathcal{L}_{MIM} = \frac{1}{|M|} \sum_{i \in M} ||\hat{X}_i - X_i||_2^2 \tag{1}$$

where M denotes the set of masked patches. We utilized a masking ratio of 60% to force the network to reconstruct subtle agricultural structures.

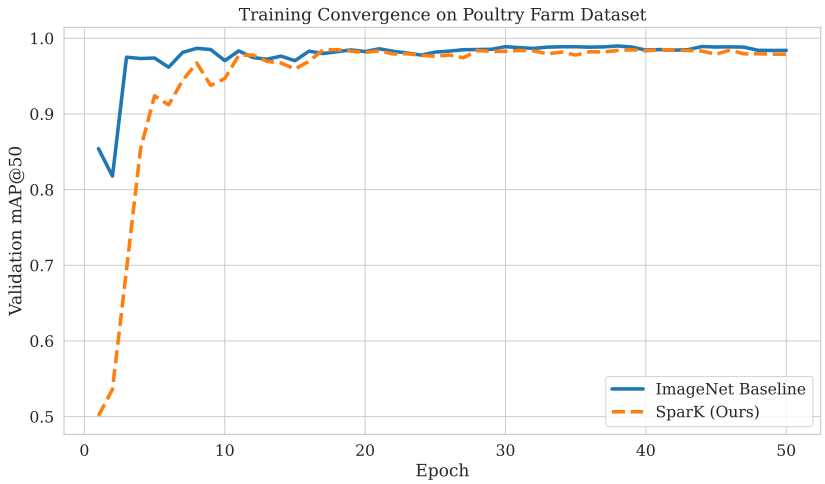


Figure 1: Training convergence metrics for the SparK pre-trained backbone during downstream fine-tuning, demonstrating rapid mAP and loss improvements.

3.2.2. Downstream Object Detection

Following pre-training, the learned convolutional backbone was injected into the YOLO11l-OBB architecture. The YOLO loss function integrates bounding box regression (using Complete Intersection over Union, CIoU), objectness confidence, and classification loss. By initializing this architecture with our SparK-pretrained weights, the network begins fine-tuning with a highly optimized agricultural visual prior.

3.3. Experimental Design

For each downstream task, an A/B experimental comparison was conducted against the official Ultralytics ImageNet pre-trained baseline. Both

Table 2: Hyperparameter configurations for downstream fine-tuning.

Task	Epochs	Optimizer	Learning Rate	Weight Decay
Poultry Farm	50	AdamW	1e-3	0.0005
GRSD	50	AdamW	1e-3	0.0005
DOTA Benchmark	20	AdamW	1e-3	0.0005

models were trained under identical hyperparameter configurations detailed in Table 2.

Additionally, a **Few-Shot Data Efficiency Ablation** was conducted on the Poultry Farm dataset, limiting training to a 10% fractional subset to assess feature robustness under extreme data scarcity.

4. Results

4.1. Downstream Transfer Parity (100% Data)

The results across all three datasets demonstrate that the SparK NAIP-pretrained backbone achieves exceptional parity with the heavily supervised ImageNet baseline.

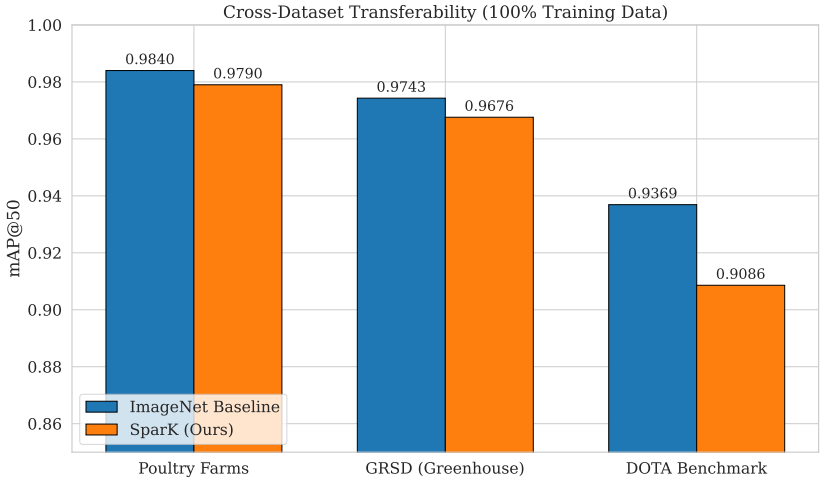


Figure 2: Multi-domain transferability comparison displaying the validation mAP@50 across three distinct downstream datasets.

As shown in Table 3, the SparK backbone achieved highly competitive state-of-the-art localization performance. On the primary poultry detection

Table 3: Comparative performance metrics for downstream tasks using 100% of the training data.

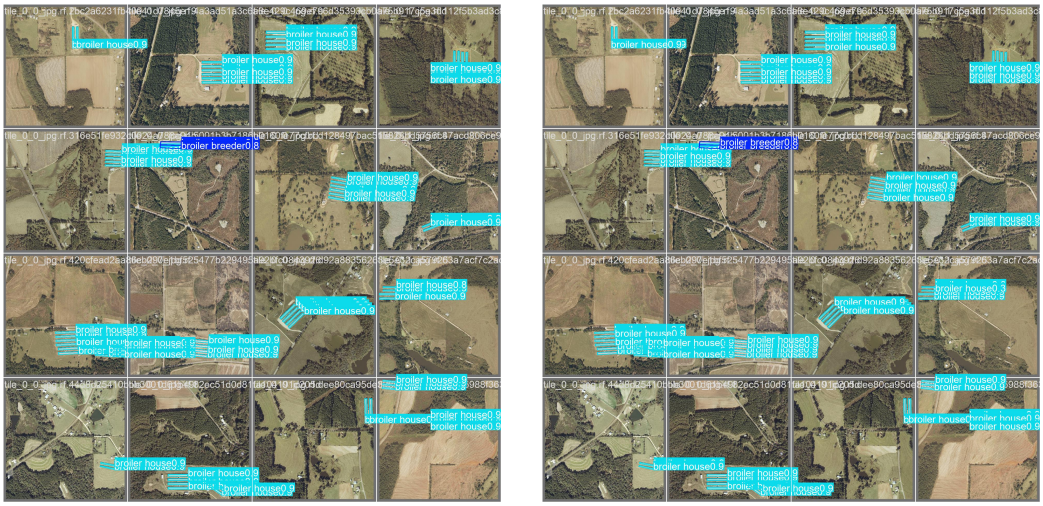
Dataset	Initialization prior	mAP50	mAP50-95	Recall
Poultry Farms	ImageNet (Supervised)	**0.9840**	**0.8421**	0.9775
	SparK NAIP (SSL)	0.9790	0.8350	**0.9831**
GRSD (Greenhouses)	ImageNet (Supervised)	**0.9743**	**0.8376**	**0.9392**
	SparK NAIP (SSL)	0.9676	0.8201	0.9210
DOTA Benchmark	ImageNet (Supervised)	**0.9369**	**0.8097**	**0.8977**
	SparK NAIP (SSL)	0.9086	0.7701	0.8564

task, while the ImageNet baseline marginally outperformed in mAP50 by 0.5%, the SparK model demonstrated superior Recall (98.3% vs. 97.7%). In agricultural surveying, higher recall is often preferred to minimize false negatives. Across the generalized GRSD and complex DOTA datasets, the SSL model trailed the ImageNet baseline by only a nominal margin (0.7% and 2.8% mAP50, respectively).

4.2. Ablation Study: Few-Shot Data Efficiency (10% Data)

To rigorously stress-test the robustness of the learned representations, we restricted the Poultry Farm training dataset to a 10% randomly sampled fraction.

Table 4 reveals a critical limitation. While the ImageNet baseline main-



(a) ImageNet Baseline (b) SparK Pre-trained (Ours)

Figure 3: Qualitative comparison of oriented bounding box (OBB) predictions on the validation set.

Table 4: Few-Shot Data Efficiency (10% of Poultry Dataset) over 50 epochs.

Data Fraction	Initialization prior	mAP50	mAP50-95	Precision
10% Few-Shot	ImageNet (Supervised)	**0.9778**	**0.7599**	**0.9722**
	SparK NAIP (SSL)	0.7064	0.5595	0.6739

tained high accuracy (97.7% mAP50) due to its massive, heavily supervised feature extraction capacity, the SparK model's performance degraded to 70.6% mAP50.

5. Discussion

The core finding of this study is the high degree of transferability achieved by self-supervised masked image modeling on unlabelled aerial data. By reaching 97.9% mAP50 on poultry farms and 96.7% on greenhouses, we provide compelling evidence that massive, explicitly labeled ground-level datasets like ImageNet are no longer a strict prerequisite for agricultural

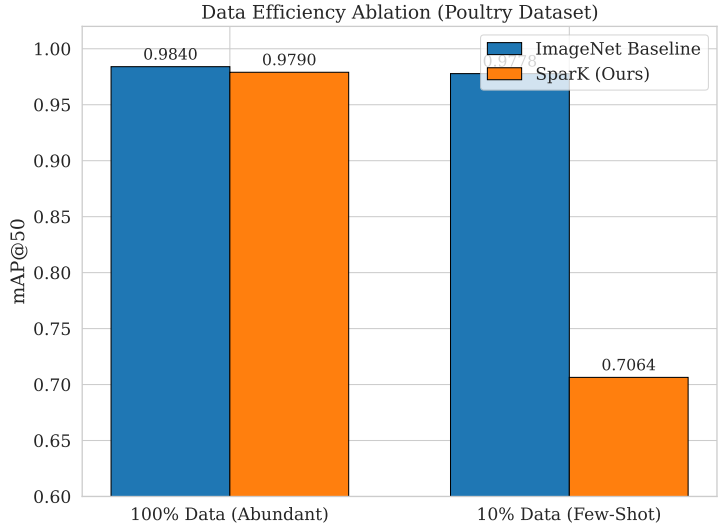


Figure 4: Data efficiency ablation study comparing the performance of ImageNet and SparK initializations on abundant (100%) versus scarce (10%) training data scenarios.

remote sensing applications.

The domain-specific SparK model learned meaningful spatial hierarchies—distinguishing synthetic materials from vegetation—purely by reconstructing masked NAIP tiles. Its superior recall on the poultry dataset suggests that its features are inherently better tuned for detecting subtle anomalies in agricultural landscapes compared to the ImageNet priors, which tend to optimize for tighter, high-confidence bounding boxes on natural scenes.

However, the 10% few-shot ablation study provides a crucial caveat. The performance collapse of the SSL model under extreme data scarcity highlights the notorious data-hunger of self-supervised learning during the pre-training phase. While the ImageNet baseline benefited from thousands of epochs over 1.2 million labelled instances, our SparK model was pre-trained for only 300 epochs. SSL models typically require highly extended pre-training schedules (e.g., 800 to 1600 epochs) to mature the feature extractors sufficiently to dominate the few-shot regime. Rather than indicating a failure of the methodology, this result highlights the vital importance of pre-training scale in self-supervised agricultural models.

6. Conclusion

This research successfully developed a robust, automated pipeline for agricultural infrastructure detection using high-resolution multispectral aerial imagery. We demonstrated that self-supervised Masked Image Modeling (SparK) on 1.5 Terabytes of unlabelled NAIP data creates a highly effective foundation model for remote sensing. The proposed domain-specific backbone achieved near-parity with a fully supervised ImageNet baseline across diverse tasks including poultry farm, greenhouse, and generalized multi-class aerial detection. While extended pre-training schedules are required to improve low-data generalization in few-shot regimes, this work establishes that leveraging abundant unannotated satellite imagery via SSL is a highly scalable and viable pathway for next-generation agricultural monitoring and precision agriculture workflows.